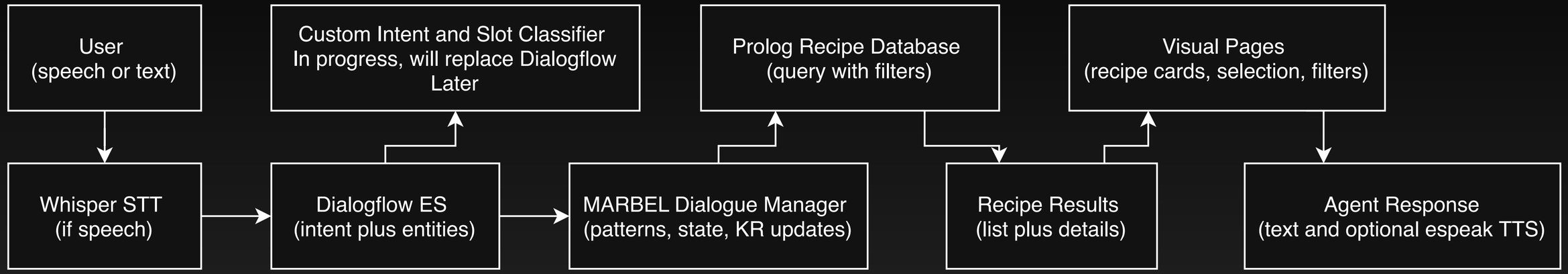
Recipe Recommendation Conversational Agent

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Recipe Recommendation Conversational Agent



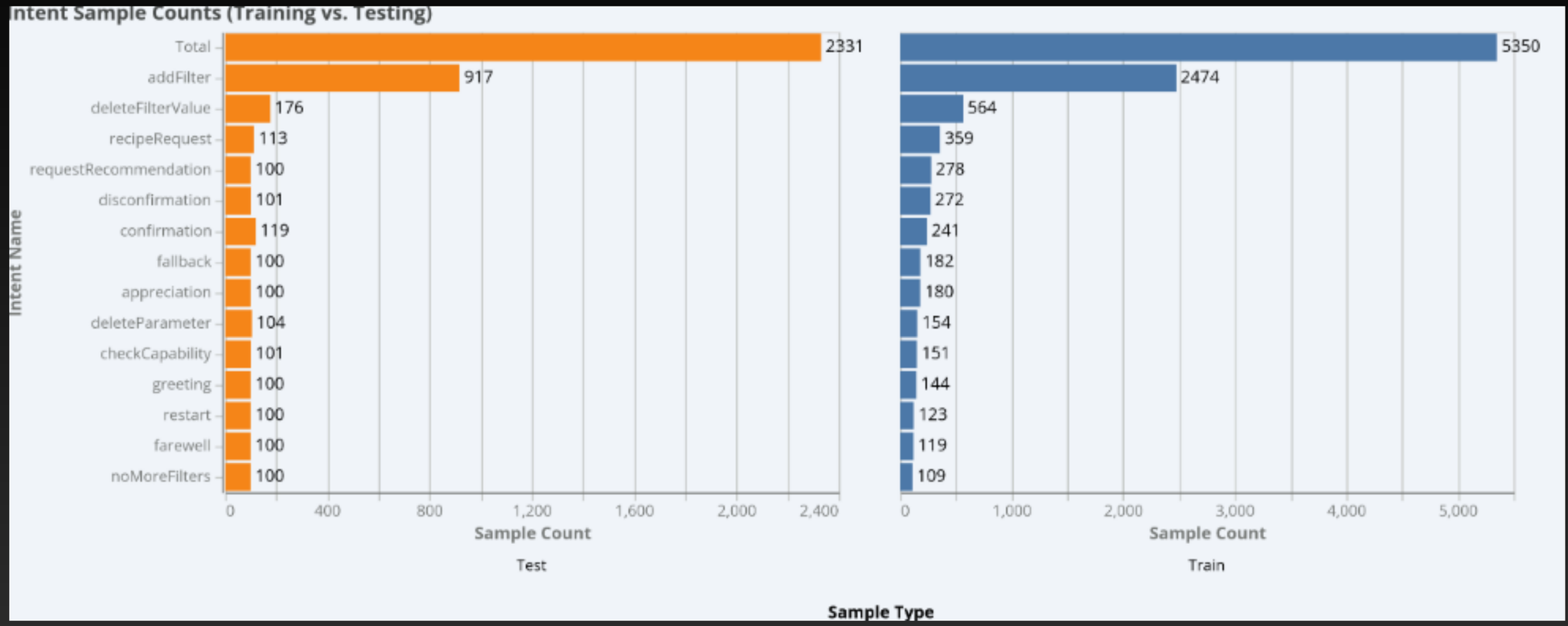
Goal: help users find recipes using preferences like cuisine, meal type, ingredients, time.

Midway scope: inclusion filtering and required dialogue flows, NLU via Dialogflow fallback.

Output: recommended recipes plus visual display and agent responses.

Custom intent and slot classifier trained and evaluated, integration planned.

Dataset Distribution Across Intents, Train vs Test



Train and test distributions are consistent (same intent ranking).

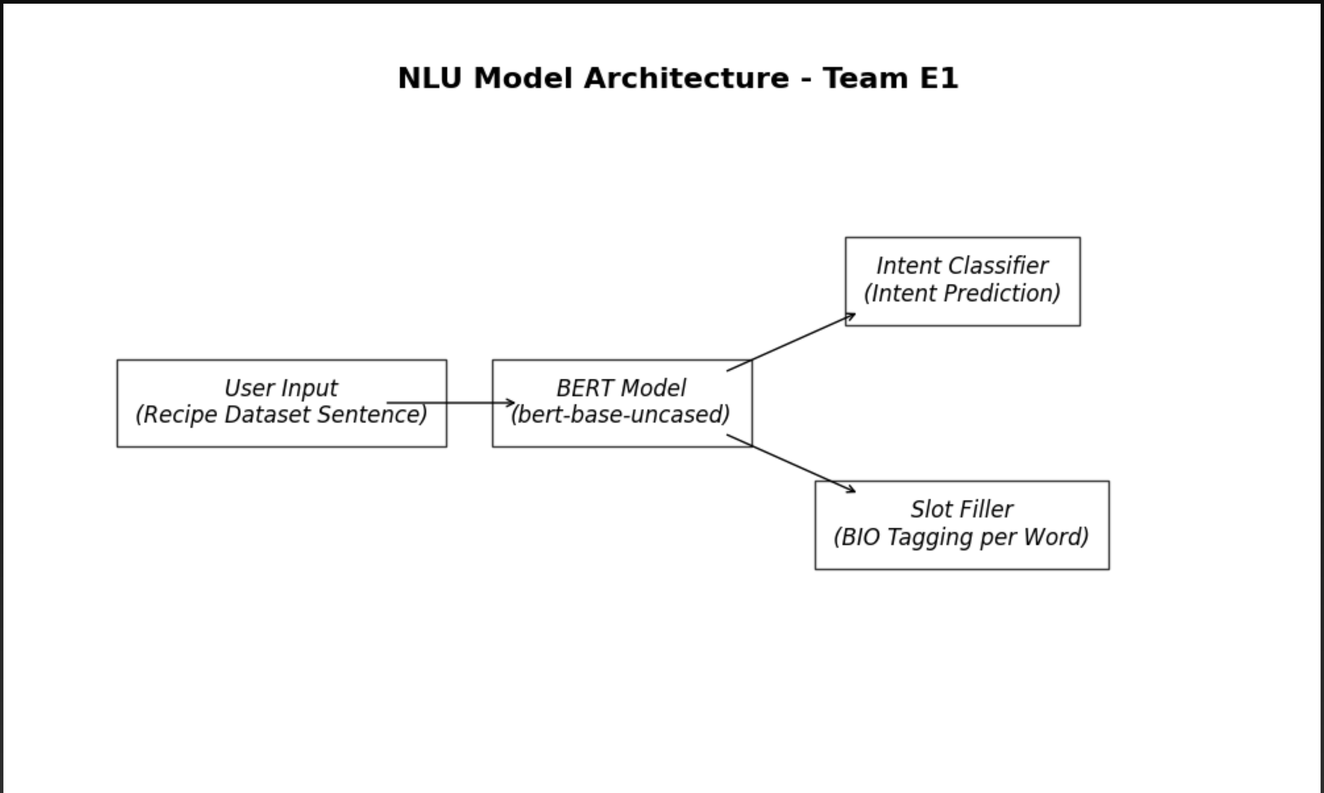
Strong class imbalance: addFilter dominates, long tail of low count intents.

Impact: report per intent metrics, not only overall accuracy.

Class imbalance motivates macro averaged metrics and per intent analysis.

NLU Model Architecture

Single shared BERT encoder with two outputs: intent classification and BIO slot tagging.



1. Shared BERT encoder produces a sentence representation and token representations.
2. Intent head predicts one intent label.
3. Slot head predicts BIO tags per token and decodes them into slots.

Evaluation Metrics vs Thresholds

Metrics:

Intent accuracy: 0.858

Intent macro precision: 0.840

Intent macro recall: 0.810

Intent macro F1: 0.800

Slot F1 score: 0.969

metric	value	threshold	pass_fail
intent_accuracy	0.8580	0.85	pass
slot_f1	0.9687	0.80	pass

Threshold check:

Intent accuracy: 0.858 vs 0.85 PASS

Slot F1: 0.969 vs 0.80 PASS

Progress and Planned Improvements

Done:

- Dialogflow imported and verified on demo phrases.
- BERT based intent and slot classifier trained and evaluated against thresholds.
- Dataset distribution and evaluation artifacts committed.

Planned improvements:

- Integrate classifier output into MARBEL routing (replace Dialogflow for final).
- Improve weak intents and rare intents with targeted data augmentation.
- Improve slot types with low performance and re evaluate.

MARBEL routing from NLU intent to action

```
1   === MARBEL TEXT DEMO (evidence generator) ===
2   Repo root: /Users/bariskaban/Desktop/VU/P-CA/agent-2026-e1
3   Latest log: /Users/bariskaban/Desktop/VU/P-CA/agent-2026-e1/artifacts/logs/marbel/marbel_text_demo_latest.txt
4   Wrote flow snippets in artifacts/logs/marbel/
5   === DONE ===
```

how evidence was generated

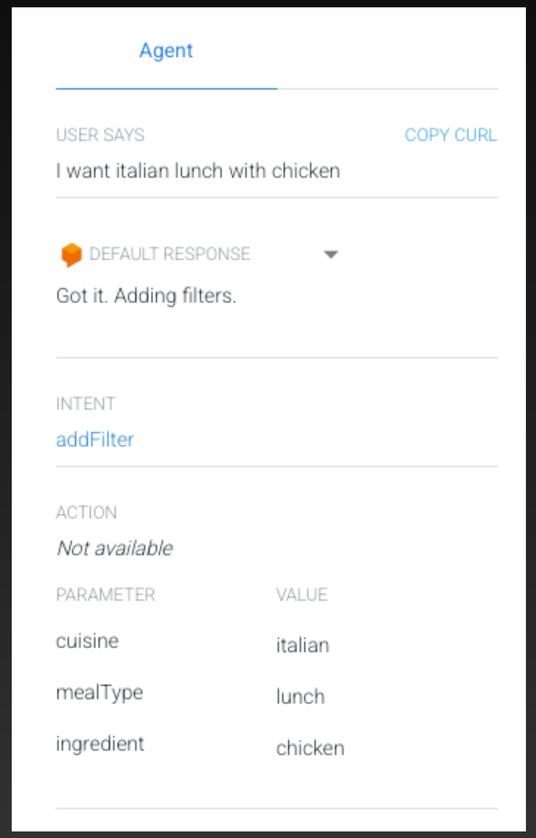
```
13  18:MARBEL_EVENT turn_start turn="3" user_text="recommend me something"
14  19:MARBEL_EVENT intent_received intent="recommend" params={}
15  20:MARBEL_EVENT pattern_chosen pattern_id="fallback_recommend"
16  21:MARBEL_EVENT kr_update changed_keys=["goal"] kr_snapshot={"goal": "recommendation"}
17  22:MARBEL_EVENT next_action agent_intent="specifyGoal" agent_text="[specifyGoal]"
18  23:MARBEL_EVENT state_transition before={"turn": 2, "active_pattern": "fallback_selfIdentification", "kr": {"goal": "recommendation"}} after={"turn": 3, "active_pattern": "fallback_recommend",
```

- intent_received
- pattern_chosen
- kr_update
- next_action
- state_transition

1. NLU provides intent and slots
2. MARBEL selects dialogue pattern and updates KR
3. KR triggers Prolog query and UI update

Inclusion Filtering Evidence

Dialogflow intent + slots (input to MARBEL)




```
24:MARBEL_EVENT turn_start turn="4" user_text="I want something with chicken"
25:MARBEL_EVENT intent_received intent="featureInquiry" params={"ingredient": "chicken"}
26:MARBEL_EVENT pattern_chosen pattern_id="fallback_featureInquiry"
27:MARBEL_EVENT kr_update changed_keys=["filters", "last_featureInquiry"] kr_snapshot={"goal": "recommendation", "filters": {"ingredient": "chicken"}, "last_featureInquiry": "I want something
28:MARBEL_EVENT next_action agent_intent="ackFilter" agent_text="[ackFilter]"
29:MARBEL_EVENT state_transition before={"turn": 3, "active_pattern": "fallback_recommend", "kr": {"goal": "recommendation", "filters": {"ingredient": "chicken"}, "last_featureInquiry": "I wan
```

Add inclusion filter (ingredient = chicken)

intent received: featureInquiry

pattern chosen: fallback_featureInquiry

KR update: filters.ingredient = chicken

next action: ackFilter

state transition: active_pattern updated to fallback_featureInquiry

```
30:MARBEL_EVENT turn_start turn="5" user_text="remove chicken"
31:MARBEL_EVENT intent_received intent="featureRemovalRequest" params={"ingredient": "chicken"}
32:MARBEL_EVENT pattern_chosen pattern_id="fallback_featureRemovalRequest"
33:MARBEL_EVENT kr_update changed_keys=["filters", "last_featureRemoval"] kr_snapshot={"goal": "recommendation", "filters": {}, "last_featureInquiry": "I want something with chicken", "last_fe
34:MARBEL_EVENT next_action agent_intent="ackFilter" agent_text="[ackFilter]"
35:MARBEL_EVENT state_transition before={"turn": 4, "active_pattern": "fallback_featureInquiry", "kr": {"goal": "recommendation", "filters": {}, "last_featureInquiry": "I want something with c
```

Remove inclusion filter (ingredient = chicken)

intent received: featureRemovalRequest

pattern chosen: fallback_featureRemovalRequest

KR update: filters cleared

next action: ackFilter

state transition: active_pattern updated to fallback_featureRemovalRequest

Live Demo Script

Step:	Step 1	Step 2	Step 3	Step 4	Step 5
User says:	hello	who are you	recommend me something	I want Italian lunch with chicken	show me option 2
Expected intent:	greeting	selfIdentify	requestRecommendation	addFilter	selectRecipe
Expected entities:	none	none	none	cuisine italian, mealType lunch, ingredient chicken	optionNumber 2
What the agent does:	greets and starts the interaction	explains what it is and what it can help with	proposes recipes or asks one narrowing question	confirms filters, updates state, refreshes results	selects the second recipe and shows details
Proof point:	greet and self identify starts correctly	self identify flow works	request recommendation flow works	inclusion filtering changes the results	selection flow works end to end

Limitations and Next Steps

Current limitations

- Midway uses Dialogflow as NLU fallback, custom classifier not yet integrated into MARBEL
- Limited robustness on rare intents and paraphrases
- End to end logging and UI screenshot evidence still being finalized (MARBEL and UI integration)

Next steps

- Integrate trained classifier into MARBEL routing and remove Dialogflow dependency for final
- Improve low performing intents and slots, re evaluate against thresholds
- Finalize full demo run script and capture end to end evidence (logs, UI updates)